

LESSON 8-4 • SESSION 2 • PRACTICE • FORM B

Representing Inequalities

Learning target – I can represent an inequality with a set of solutions and describe them. • 6.EE.B.8

REMEMBER

- **Closed** (filled) circle: the boundary counts ($\leq$, $\geq$).
Open circle: it does not ($<$, $>$).
- Arrow **left** for less than. Arrow **right** for greater than.
- Check the story: people and tickets come only in **whole numbers**.

WORKED EXAMPLE

An elevator allows no more than 8 people. Show the solutions for p .**Step 1** Write the rule: $p \leq 8$.**Step 2** "No more than" includes 8, so draw a closed circle at 8 and an arrow left.**Step 3** People come in whole numbers: 0, 1, 2, ..., 8.**Answer** $p \leq 8$, whole numbers 0 to 8**WORDS** number line (*recta numérica*) • open circle (*círculo abierto*) • closed circle (*círculo cerrado*)

PRACTICE

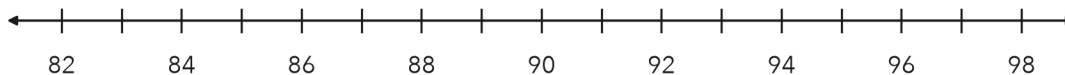
show your work • label units

1. Write the inequality for each graph. Use x .

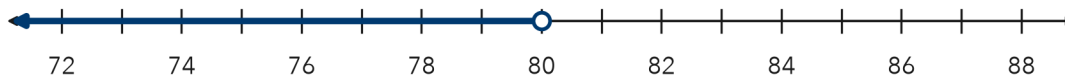
a. open circle at 2, arrow left _____

b. closed circle at -3 , arrow right _____

c. closed circle at 5, arrow left _____

d. open circle at -1 , arrow right _____2. **Meeting Room.** A meeting room may hold at most 90 people at one time.a. Write an inequality for n , the number of people. _____

b. Graph it on the number line.

3. **Fish Tank.** The graph shows the safe water temperatures, t , for a fish tank. Write the inequality. Is 80 degrees a solution?**Hint** Look at the circle at 80. Is it open or closed?**Answer** _____4. **Locker Bag.** A bag must be shorter than 24 inches to fit in a locker. Write an inequality for h . Name two solutions and one non-solution.

Show your work.

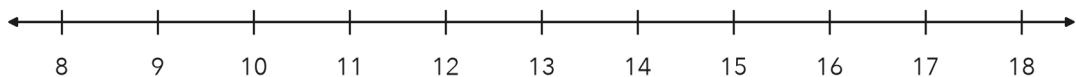
Answer _____5. **Lab Station.** A lab station may hold at most 6 students. Write an inequality for s . Is 6 a solution?**Hint** "At most" includes the boundary.

Show your work.

Answer _____

6. **Parade Float.** A float must be shorter than 13 feet to pass under a bridge.

Hint "Shorter than" leaves the boundary out.



- a. Write an inequality for h . _____
- b. Graph it on the number line.

7. **Name Solutions.** For $n > 4$, name three solutions and one number that is not a solution.

Show your work.

Answer _____

8. **Delivery Limit.** A van's load, w , must be at least 40 pounds and at most 800 pounds.

- a. Write both inequalities. _____
- b. Name three loads that fit both rules. _____
- c. Why is 800 pounds allowed but 801 pounds is not? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Theo's Work.** A study hall may seat at most 32 students. Theo wrote $s < 32$. Explain his mistake. Write the correct inequality.

TEST PRACTICE

choose the best answer

10. Which value is a solution of $n \leq 9$?

- A** 9 **B** 9.5 **C** 10 **D** 12

11. Which value is NOT a solution of $x \leq 7$?

- A** 7 **B** 6.5 **C** 0 **D** 7.5

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

n < 90 Q	t ≤ 80 Z	s < 6 X	w ≥ 40 S	h < 13 W	w > 40 J	s ≤ 6 O	h < 24 R	n ≤ 90 A	t < 80 R
		#2a	#3	#4	#5	#6a	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____